Week 1 Introduction

Welcome to DSC 670 - Advanced Uses of Generative AI!

Popular culture, including media outlets, has painted a picture of generative AI that is exciting, promising, misleading (at times), and filled with hyperbole that's...*speculative*. In Week 1, you begin to learn some of the facts related to generative AI, with an ample amount of opportunity to speculate on your own.

People often speak about "ChatGPT" as if it's a generative AI model, but it's not a model, but an *interface* into a model. As such, it's not a direct feed to/from one or more generative AI models, but a kind of orchestrator and filter. It's also a means for the model maker (OpenAI) to gather more data from you, which may be used to make the models better over time.

But there's a catch: whatever you provide to an interface belongs to the interface maker, so be careful. This week, you'll learn about what the interfaces are, how they work, and how you can use them without unintentionally giving the interface makers things you don't want them to have. You'll also get your first glimpse into how to use generative AI via prompting.

While there are times the first week can be easier than others, this is **not** the case with this class. There's a lot to read, along with a two part exercise, in addition to the discussion board. Stay on top of the week.

Contents of the Week:

- Introduction
- Readings and Videos
- Week 1 Discussion/Participation
- Week 1 Exercise
- Review the Final Project Instructions

Objectives/Topics:

- Establish the basic generative AI vocabulary.
- Define GAI in the context of artificial intelligence.
- Provide a brief history of GAI.
- Identify some of the major GAI models and their makers.
- Describe specific GAI capabilities.
- Describe some notable GAI use cases in the real world.

Helpful Resources:

Before you begin this week, here are some recommended resources you'll need going forward.

- GitHub (https://github.com/) – A great resource to share the content you are creating with potential employers and connect with other data scientists in the community. We will be using this in our course, but it is also a great resource for you to continue using outside of

your courses for any skunk work projects you are working on and for showing the world your expertise.

- Medium (https://medium.com/)– Aggregates articles on virtually any topic you can imagine, particularly data science. This is best to read daily.